

Supplementary Tables

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Title: P3: Persistence-Aware Admission Control for DoS-Resilient BLE Resolvable Private Address Resolution

The tables below report deterministic simulator outputs. They do not constitute measured BLE controller timing, packet-loss, energy, or production-firmware evidence. The accompanying reviewer artifact archive contains the seed-level summaries, configurations, manifests, audit tables, and reproduction scripts.

Supplementary Table S1

Supplementary Table S1. Complete long-window validation results for N = 512 and RL = 8 (1800-second runs; 300-second warmup; n = 3 seed means for each scenario/method pair).

Scenario	Method	n	Attack/min	Attack AES/min	Legit rate	False defer
heavy_flood	BudgetDoS	3	10000	45192.00	0.5233	0.4767
heavy_flood	P3-NoPersist	3	10000	45198.72	0.5602	0.4403
heavy_flood	P3-NoBudget	3	10000	5035658.88	1.0000	0.0088
heavy_flood	P3-NoCache	3	10000	45192.00	0.5233	0.4774
heavy_flood	StaticRL	3	10000	5035658.88	1.0000	0.0000
heavy_flood	ZephyrCache	3	10000	5035658.88	1.0000	0.0000
medium_flood	BudgetDoS	3	1000	43874.88	0.5809	0.4191
medium_flood	P3-NoPersist	3	1000	44593.92	0.7554	0.2459
medium_flood	P3-NoBudget	3	1000	502353.60	1.0000	0.0072
medium_flood	P3-NoCache	3	1000	43874.88	0.5809	0.4211
medium_flood	StaticRL	3	1000	502353.60	1.0000	0.0000
medium_flood	ZephyrCache	3	1000	502353.60	1.0000	0.0000
normal_capacity	BudgetDoS	3	0	0.00	0.8814	0.1186
normal_capacity	P3-NoPersist	3	0	0.00	0.9855	0.0234
normal_capacity	P3-NoBudget	3	0	0.00	1.0000	0.0096
normal_capacity	P3-NoCache	3	0	0.00	0.8814	0.1388
normal_capacity	StaticRL	3	0	0.00	1.0000	0.0000
normal_capacity	ZephyrCache	3	0	0.00	1.0000	0.0000

Supplementary Table S2

Supplementary Table S2. Paired modeled-observable privacy audit (20 paired seeds per load/filter condition). The external proxy contains defer rate, mean modeled delay, and p95 modeled delay.

Load	Δ (s)	Method	Pairs	Initial audit	External-proxy AUC	Bootstrap 95% CI
no_attack	0	P3-Persist	20	All matched	1.000000	1.000000–1.000000
no_attack	0	P3-NoPersist	20	All matched	1.000000	1.000000–1.000000
no_attack	0	BudgetDoS	20	All matched	0.500000	0.500000–0.500000
no_attack	60	P3-Persist	20	All matched	0.500000	0.500000–0.500000
no_attack	60	P3-NoPersist	20	All matched	0.500000	0.500000–0.500000
no_attack	60	BudgetDoS	20	All matched	0.500000	0.500000–0.500000
medium_flood	0	P3-Persist	20	All matched	1.000000	1.000000–1.000000
medium_flood	0	P3-NoPersist	20	All matched	1.000000	1.000000–1.000000
medium_flood	0	BudgetDoS	20	All matched	0.500000	0.500000–0.500000
medium_flood	60	P3-Persist	20	All matched	0.500000	0.500000–0.500000
medium_flood	60	P3-NoPersist	20	All matched	0.500000	0.500000–0.500000
medium_flood	60	BudgetDoS	20	All matched	0.500000	0.500000–0.500000
heavy_flood	0	P3-Persist	20	All matched	0.500000	0.500000–0.500000
heavy_flood	0	P3-NoPersist	20	All matched	0.500000	0.500000–0.500000
heavy_flood	0	BudgetDoS	20	All matched	0.500000	0.500000–0.500000
heavy_flood	60	P3-Persist	20	All matched	0.500000	0.500000–0.500000
heavy_flood	60	P3-NoPersist	20	All matched	0.500000	0.500000–0.500000
heavy_flood	60	BudgetDoS	20	All matched	0.500000	0.500000–0.500000

Note: Every paired job used identical visible timestamp, RSSI, payload-length, RPA-value, repetition-pattern, load, and duplicate-filter inputs; only the offline identity label and normal resolution outcome differed. ‘All matched’ refers only to the tested initial pre-resolution decision. AUC values are simulator-derived modeled-proxy results and do not establish real end-to-end side-channel freedom. Full paired effects and internal mechanism diagnostics are supplied as CSV artifacts.

Supplementary Table S3

Supplementary Table S3. P3-Persist joint RPA-epoch, modeled observation-loss, and benign-unknown boundary (20 seeds per cell).

Cell	Epoch (min)	Loss	Benign/min	Benign mode	Offered legit	Observed legit	Observed false defer	Attack AES/min
main	1	0.0	100	unique	0.705290	0.705290	0.300297	49,655.088
main	1	0.0	1000	unique	0.706148	0.706148	0.299293	45,647.280
main	1	0.0	5000	unique	0.707047	0.707047	0.298758	33,449.472
main	1	0.2	100	unique	0.553429	0.689848	0.315029	49,688.352
main	1	0.2	1000	unique	0.550571	0.689817	0.315005	45,647.280
main	1	0.2	5000	unique	0.552805	0.690178	0.315520	33,413.184
main	1	0.4	100	unique	0.400575	0.668126	0.336391	49,645.008
main	1	0.4	1000	unique	0.405555	0.673816	0.331320	45,566.640
main	1	0.4	5000	unique	0.402741	0.671334	0.333547	33,477.696
main	5	0.0	100	unique	0.853841	0.853841	0.153106	49,708.512
main	5	0.0	1000	unique	0.853017	0.853017	0.153234	45,707.760
main	5	0.0	5000	unique	0.854705	0.854705	0.152061	33,468.624
main	5	0.2	100	unique	0.664156	0.832467	0.174574	49,724.640
main	5	0.2	1000	unique	0.664427	0.831558	0.175253	45,674.496
main	5	0.2	5000	unique	0.664345	0.832450	0.174659	33,453.504
main	5	0.4	100	unique	0.482583	0.802174	0.205532	49,714.560
main	5	0.4	1000	unique	0.486090	0.802804	0.204807	45,687.600
main	5	0.4	5000	unique	0.482278	0.803271	0.204016	33,516.000
main	15	0.0	100	unique	0.955313	0.955313	0.050653	49,783.104
main	15	0.0	1000	unique	0.955667	0.955667	0.049741	45,765.216
main	15	0.0	5000	unique	0.955484	0.955484	0.049787	33,497.856
main	15	0.2	100	unique	0.756427	0.943452	0.062871	49,778.064
main	15	0.2	1000	unique	0.754612	0.943871	0.061772	45,754.128
main	15	0.2	5000	unique	0.755108	0.943666	0.061952	33,430.320
main	15	0.4	100	unique	0.557582	0.926261	0.079763	49,792.176
main	15	0.4	1000	unique	0.556241	0.925610	0.079613	45,778.320
main	15	0.4	5000	unique	0.555045	0.926269	0.079337	33,287.184
control: no_attack_baseline	15	0.0	100	unique	0.988384	0.988384	0.018666	0.000
control: no_attack_worst	1	0.4	5000	unique	0.406062	0.674463	0.330103	0.000
control: repeated_benign_worst	1	0.4	5000	repeated	0.345137	0.575279	0.425083	33,466.608

Note: The 27 main cells use 10000 attack RPAs/min and unique benign addresses. The three controls are the no-attack baseline, the no-attack adverse boundary, and the attacked repeated-benign adverse boundary. Observation loss is applied by the simulator after event generation and before scheduling; it is a controlled proxy for missed observations, not measured BLE packet loss. Generated-but-unobserved legitimate events remain in the offered denominator.

Supplementary Table S4

Supplementary Table S4. Twenty-seed confirmation of the preregistered adaptive local-search union.

Δ (s)	Rank	Strategy	n	Attack AES/min	95% CI	Mean/cap	Legit rate	False defer	Removable fraction
0	1	pool 224, r=1, p=120	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	2	pool 256, r=1, p=105	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	3	pool 256, r=1, p=120	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	4	pool 256, r=1, p=135	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	5	pool 256, r=2, p=120	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	6	pool 320, r=1, p=105	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	7	pool 320, r=1, p=120	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	8	pool 320, r=1, p=135	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	9	pool 320, r=1, p=75	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	10	pool 320, r=1, p=90	20	146,162.016	145,986.621–146,337.411	0.966680	0.640442	0.360126	0.000000
0	11	unique	20	45,765.216	45,606.392–45,924.040	0.302680	0.955667	0.049741	0.000000
60	1	pool 320, r=1, p=75	20	146,196.288	146,018.171–146,374.405	0.966907	0.674902	0.325627	0.970628
60	2	pool 320, r=1, p=135	20	146,195.280	146,015.168–146,375.392	0.966900	0.675190	0.325339	0.970628
60	3	pool 320, r=1, p=90	20	146,193.264	146,014.002–146,372.526	0.966887	0.674165	0.326364	0.970629
60	4	pool 320, r=1, p=105	20	146,192.256	146,012.078–146,372.434	0.966880	0.673834	0.326695	0.970628
60	5	pool 320, r=1, p=120	20	146,192.256	146,012.078–146,372.434	0.966880	0.673232	0.327297	0.970628
60	6	pool 256, r=2, p=120	20	129,012.912	128,998.080–129,027.744	0.853260	0.929699	0.075611	0.974394
60	7	pool 256, r=1, p=105	20	129,001.824	128,989.618–129,014.030	0.853187	0.928646	0.076663	0.974396
60	8	pool 256, r=1, p=135	20	128,980.656	128,961.472–128,999.840	0.853047	0.928646	0.076663	0.974401
60	9	pool 256, r=1, p=120	20	128,962.512	128,939.305–128,985.719	0.852927	0.928646	0.076663	0.974403
60	10	pool 224, r=1, p=120	20	112,821.408	112,799.312–112,843.504	0.746173	0.928646	0.076663	0.977607
60	11	unique	20	45,758.160	45,597.956–45,918.364	0.302633	0.928646	0.076663	0.000000

Note: The confirmation union contains the mechanically selected top five, the historical reference, five single-axis neighbors, and the unique-address baseline after deduplication. The complete 168-strategy, five-seed screening table is supplied as `adaptive_local_screen_complete.csv`. Overlapping confidence intervals identify a tied damaging local region; the search does not establish a global or unique attacker optimum. The cap is 151,200 AES-equivalent attempts/min for the stated modeled parameters.